Rowan University

Electrical & Computer Engineering

Engineering Clinics

Project Title: Asset Predictive Maintenance

Project Team

PM Faculty Name: Dr. Yashwant Sinha

PM Department: ME

Sponsor (if applicable) :

Date Submitted: 12/17/2025

Project Students - Name, Dept, Banner ID #, and Signature

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Use a second cover page, if project included more than five students

Table of Contents

	Report Component	Pages	Weight
Engineering Design Process (Graded as team based on the report)	**Preliminaries**		
	Abstract	1	5
	Introduction & problem statement	1	
	Technical requirements, specifications, and deliverables, timeline (Gannt chart)	4	
	References and acknowledgements	12	
	Constraints	3	5
	Applicable Engineering Standards	4	5
	Approach, Design and Validation: Solution of Complex Eng. Problems & Engineering Judgment		
	Background and main approach	2	45
	Trade-offs,alternative solutions, iterative design, risks & opportunities for each	3	
	Design and implementation of experiments, analysis of results and validation	5	
	Evidence of appropriate design that meets specified requirements / constraints	3	
	Discussions and conclusions - Use engineering judgment to to draw conclusions	12	
	Engineering Design Process – Project Team Grade from This Report	**0**	**Out of 60**
Impact & Prof. Considerations (Graded individually based on IPCS)◆	**Impact Factors** (Enter consolidated answers in the report, submit separate IPCS for each student)		
	Public health, welfare, safety	11	Graded on IPCS (12)
	Global, social, and cultural factors	11	
	Environmental factors	11	
	Economic factors	11	
	Professional and Ethical Issues in Consideration of The Above-stated Factors	12	(10)
	Teamwork, Lessons & New Knowledge Learned		
	Team members and their respective involvements	4	(4)
	Main takeaways, lessons, new knowledge learned	11	(4)
	Effective Communication (from poster presentations)		(10)
	Individual Considerations – Graded individually on IPCS		**40 %**
	Total		**100 %**

★ With their signature, each student affirms that they have worked on the project equally and measurably in the areas indicated under Teamwork, that they properly represent their work as their own and that they have properly cited the source of the content obtained from other sources.

◆ For all individual consideration topics, enter project wide answers in this report (and cite the page numbers here). However, your grade will come from the Impact & Professional Considerations Statement (IPCS).

All grading and evaluations are for faculty use only. Students enter other project information and page numbers as indicated above.

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All grading and evaluations are for faculty use only. Students enter other project information and page numbers as indicated above.

Grading Components and Weights

The clinic grade will consist of two components: an engineering design process grade (60%) that is given to the entire team, and an impact and professional considerations grade (40%) provided separately to each student based on the Impact and Professional Considerations Statement (IPCS). Your final grade will be based on the sum of the two. The breakdown of these two grade components is provided below. Note that – if in the judgement of the project manager / faculty, a student participates / contributes measurably less than other team members – that student's engineering design process grade may also be lower than the team grade.

For each category, you will be graded on a four-point scale rubric: Excellent (100 points), Proficient (75 points), Developing (50 points) and Unsatisfactory (0 points). These points will be multiplied by the category weights given on the cover page and below (in parentheses). Detailed descriptions and definitions are available on the grading rubric.

Preliminaries (5): Follows proper technical reporting process with clear definition and implementation of problem statement / definition; technical specs / requirements; timeline / tasks / deliverables; references and proper formal technical writing language;

Multiple Realistic Constraints (5) – must include multiple constraints, with reasoning, that limit your design, solution. Time and money – without sufficient justifications – cannot be the sole constraints.

Applicable Engineering Standards (5) – must include the published applicable standard, and how / where that standard was used in the design. Must cite the standard.

Engineering Design Process – Approach, Design and Validation (45 points) Solution of complex engineering problems and using sound engineering judgment – must produce and articulate a design that meets the specified requirements and constraints, show analysis and synthesis with risks and trade-offs of multiple solutions and alternatives for an iterative design process, appropriate experiments, evaluation and validation to determine whether the design meets the requirements. Draw meaningful conclusions based on sound engineering judgment.

Impact factors (12 points) – must discuss – either in the Approach / Design or Discussion or in a separate section, the specific design decisions **you** have taken in consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors (these are connecting with "AND", meaning they all need to be considered).

Effective Communications (10 points) – must demonstrate effective communication skills with a range of audiences, using oral, written and/or poster formats. The discussions in each medium must have a suitable level of formal language and technical rigor for the intended audience. The oral presentation must be fluent and articulate. The report must follow established style guidelines.

Ethical and Professional Responsibilities (10 points) - Discuss what ethical issues may arise (or have risen) in the context of global, economic, environmental, societal factors, and how they can be addressed.

Effective Functioning on a Team (4 points) - Explain the role of each individual in the project, and how did s/he contribute to the team (leadership, establishing goals, planning, creating inclusive env., etc.). Individual contributions of each team member must be verified by the project manager and other team members.

New Knowledge and Lifelong Learning (4 Points) - A detailed explanation on what new knowledge and skills were gained, the mechanism for acquiring such knowledge/skills, and how were they applied to the project.

Predictive Maintenance of Physical Assets: Application to Wind Turbines

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***Abstract*—**This project develops a predictive maintenance framework for wind turbines using condition-based monitoring, IoT infrastructure, and data-driven analysis. The objective is to integrate sensor data and SCADA indicators with analytical and AI-based methods to identify early failure signatures in critical turbine subsystems. An IoT architecture is implemented using embedded sensors, MQTT communication, and cloud-based processing. Quantitative analysis employs statistical outlier detection and physics-based relationships, while qualitative analysis and FMEA are used to prioritize failure risks. Results highlight the rotor and gearbox as high-risk components and demonstrate improved fault detection compared to preventive maintenance. Future work includes turbine-specific model tuning and expanded datasets for increased prediction accuracy.

I. Introduction

Wind energy has become a critical component of the global transition toward sustainable and renewable power generation. Modern wind turbines operate under highly variable mechanical, electrical, and environmental conditions, leading to gradual component degradation over time. Traditionally, maintenance strategies for wind turbines have relied on preventive or reactive approaches, which often result in unnecessary servicing or unexpected failures. Recent advances in sensor technology, data analytics, and artificial intelligence have enabled predictive maintenance methods that leverage real-time operational data to assess equipment health and anticipate failures before they occur.

A. Problem Statement

Despite widespread deployment of sensors and SCADA systems in wind turbines, much of the collected data is underutilized for proactive maintenance decision-making. Existing maintenance strategies often fail to detect early-stage faults, particularly in high-risk components such as the rotor system, gearbox, and control mechanisms. This limitation results in unplanned downtime, increased repair costs, and reduced energy production. There is a need for an integrated predictive maintenance framework that effectively combines sensor data, data analytics, and risk assessment to improve failure detection and maintenance planning.

B. Project Significance

Implementing predictive maintenance for wind turbines offers significant technical and economic advantages. Early fault detection can reduce catastrophic failures, extend component lifespan, and optimize maintenance schedules. By integrating IoT-based data acquisition, quantitative and qualitative analysis, and Failure Mode and Effects Analysis (FMEA), this project demonstrates a systematic approach to identifying high-risk components and prioritizing maintenance actions. The proposed framework contributes to improved turbine reliability, reduced operational costs, and increased energy availability, supporting the long-term sustainability of renewable energy infrastructure.

C. Scope and Limitations

This project focuses on the development of a conceptual predictive maintenance system for wind turbines using sensor-based condition monitoring and SCADA data analysis. The scope includes vibration, temperature, and operational parameter monitoring, cloud-based data processing, and risk prioritization through analytical methods. However, the study is limited by the availability of historical turbine-specific data and does not incorporate real-time deployment on an operating wind farm. Additionally, environmental and site-specific factors are generalized rather than customized for individual turbine locations.

D. Expected Outcomes

The expected outcome of this project is a structured predictive maintenance framework capable of identifying early warning indicators and high-risk turbine components. The system is anticipated to improve fault detection accuracy compared to traditional preventive maintenance approaches and provide actionable insights for maintenance planning. Ultimately, this work aims to demonstrate the feasibility of integrating sensor data, IoT infrastructure, and analytical models to support data-driven maintenance strategies for wind turbine applications.

II. Literature Review

The growing reliance on wind energy as a major renewable power source has increased the need for advanced maintenance strategies to ensure system reliability and cost efficiency. Wind turbines operate under highly variable mechanical and environmental conditions, making critical components such as gearboxes, bearings, blades, and control systems susceptible to progressive degradation and unexpected failure [3], [4]. Traditional preventive maintenance approaches rely on fixed service intervals and often result in unnecessary maintenance actions or unplanned downtime. Recent studies identify predictive maintenance as a data-driven alternative that leverages sensor measurements, SCADA data, and machine learning techniques to detect early fault indicators and estimate remaining useful life [2]. By integrating IoT-enabled condition monitoring with statistical analysis and failure-mode prioritization, predictive maintenance frameworks have demonstrated significant reductions in operational costs, improved fault detection accuracy, and extended asset lifespan in wind turbine applications [1].

A. Technical Theory

Predictive maintenance in wind turbine systems is grounded in condition-based monitoring and data-driven modeling, where equipment health is inferred from continuous sensor and SCADA data rather than fixed maintenance intervals. Machine learning–based anomaly detection models assume that normal turbine operation follows stable multi-variable patterns, and deviations from these patterns indicate early stages of component degradation. Techniques such as autoencoders are commonly applied to identify abnormal operating behavior in vibration, temperature, and power signals, with Autoencoders demonstrating strong performance due to their ability to capture nonlinear system dynamics [5].

Prognostic models extend anomaly detection by estimating the remaining useful life (RUL) of critical components, including gearboxes and bearings. Long Short-Term Memory (LSTM) neural networks are particularly effective for this task, as they model temporal dependencies and degradation trends in sequential sensor data, enabling accurate failure forecasting hours to months in advance [5]. Complementary SCADA-based predictive maintenance frameworks integrate multivariate statistical control methods with machine learning to detect incipient faults well before catastrophic failure, improving maintenance planning and reducing downtime [6]. These models collectively support a proactive and scalable predictive maintenance strategy applicable to complex industrial assets, including wind turbines and large-scale renewable energy systems [7].

B. Applications

Predictive maintenance systems developed in this project have practical applications in wind farm operations by enabling continuous health monitoring and early fault detection of critical turbine components such as gearboxes, bearings, and generators [2]. SCADA-based predictive models can be deployed to track operational parameters including temperature, vibration, and power output, allowing abnormal trends to be identified well before catastrophic failure occurs [4].

Machine learning–driven prognostic models further support maintenance planning by estimating remaining useful life and prioritizing high-risk components across large turbine fleets [1]. By shifting maintenance strategies from reactive or time-based approaches to condition-based decision-making, wind farm operators can reduce unplanned downtime and optimize maintenance scheduling [5]. Beyond wind energy, similar predictive maintenance frameworks have been successfully applied to manufacturing systems and other industrial assets, demonstrating scalability and cross-industry applicability [3].

C. Why this technique

Predictive maintenance was chosen over traditional preventive maintenance in this project because it leverages real-time data and advanced analytics to assess the actual condition of turbine components rather than relying on fixed service intervals. Preventive maintenance schedules tasks based on historical averages and calendar or usage

thresholds, which can result in unnecessary interventions or missed faults between inspections [8]. In contrast, predictive maintenance continuously evaluates sensor and SCADA data to forecast when failures are likely to occur, enabling maintenance to be performed just in time and thus reducing unplanned downtime and optimizing resource allocation [8]. This data-driven approach, supported by machine learning and IoT integration, improves asset reliability, extends equipment life, and enhances operational efficiency when applied to complex systems like wind turbines [8].

III. Project Objectives

The main objective of this project is to develop and demonstrate a predictive maintenance approach for wind turbines that enables early fault detection and supports data-driven maintenance decisions.

A. Project Goals

- Develop a condition-based maintenance framework for wind turbines using sensor measurements and SCADA data.
- Establish an IoT-based system for reliable collection, transmission, and storage of real-time operational data through embedded hardware and cloud services.
- Utilize quantitative analysis methods, including statistical outlier detection and physics-based relationships, to detect early signs of component degradation.
- Conduct qualitative analysis and apply Failure Mode and Effects Analysis (FMEA) to identify and prioritize high-risk components and failure mechanisms.
- Combine analytical insights into an integrated predictive model that connects component health assessment with failure risk and recommended maintenance actions.

B. Success Criteria & Constraints

- Anomaly Detection Accuracy: Achieves high precision and recalls in identifying abnormal operating behavior, with performance comparable to benchmark machine learning models.
- Prediction Lead Time: Detects failure indicators sufficiently early (hours to weeks in advance) to enable planned maintenance interventions.
- System Reliability: Demonstrates consistent data transmission, processing, and analysis without significant data loss or latency.
- Risk Prioritization Effectiveness: Successfully ranks turbine components by Risk Priority Number (RPN) using FMEA to support maintenance decision-making.
- Industry Alignment: Ensures the framework aligns with condition-based maintenance practices commonly used in wind energy and industrial asset management.

IV. Methodology

This project aims to design, develop, and evaluate a predictive maintenance system for wind turbines using sensor-based condition monitoring, SCADA data, and data-driven analytics. The focus is to assess the effectiveness of integrating IoT infrastructure, anomaly detection techniques, and risk prioritization methods to identify early failure indicators, optimize maintenance decisions, and improve the reliability and efficiency of wind turbine operations.

A. Tools, Software, Materials

This project utilized a combination of hardware, software, and cloud-based tools to implement the predictive maintenance framework. Hardware components included vibration and temperature sensors, along with an ESP32 microcontroller for data acquisition and wireless communication. Software tools included Python for data processing and anomaly detection, MATLAB for relationship analysis and visualization, and Microsoft Excel for organizing and storing sensor datasets. Cloud services were implemented using AWS IoT Core for device communication, AWS Lambda for backend processing, and DynamoDB for structured data storage. These tools collectively enabled real-time data collection, processing, and analysis.

B. Experimental Design Setup

The experimental design focused on simulating a wind turbine condition-monitoring environment using sensor data representative of turbine operation. Sensors were configured to measure parameters such as vibration intensity and temperature trends associated with common failure modes. Sensor outputs were transmitted via the ESP32 microcontroller to the AWS IoT platform, where data was stored and processed. Historical and simulated datasets were then analyzed to establish baseline operating conditions and identify abnormal behavior indicative of component degradation.

C. Algorithms, Formulas, System architecture

The system architecture consists of four primary layers: sensor data acquisition, IoT communication, data processing, and predictive analytics. Quantitative analysis employed statistical outlier detection using the Interquartile Range (IQR) method to flag abnormal sensor readings. Physics-based relationships between SCADA parameters were analyzed to detect deviations from expected operational behavior. Qualitative analysis and Failure Mode and Effects Analysis (FMEA) were used to contextualize analytical findings and prioritize component risks. These methods were integrated into a unified predictive model.

D. Data Collection Procedures

Sensor and SCADA data were collected in structured formats and stored as CSV files for analysis. Data collection involved continuous sampling of vibration and temperature parameters over defined intervals. Pre-processing steps included data cleaning, normalization, and filtering to remove incomplete or irrelevant entries. This process ensured data consistency and reliability prior to analysis and modeling.

E. Work Distribution

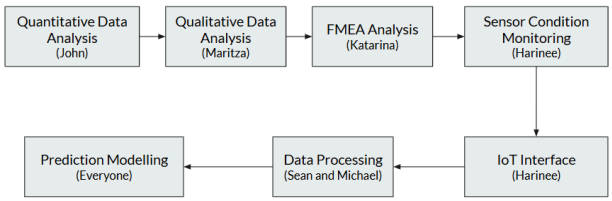


Figure 1: Distribution of Roles

Table 1. Project Design Workflow

Date	**Milestone**
Week 2 (9/8-9/12)	Group Members are assigned individual tasks.
Week 3 (9/15-9/19)	Regroup to refine assignments and review others' work
Week 4 / Week 5 (9/22-9/26, 9/29-10/3)	Quantitative analysis done and data fully cleaned
Week 6 (10/6-10/10)	Code begins to be developed and outputs are discussed as a team for desired results
Week 7 (10/13-10/17)	Qualitative analysis data is found and organized into component breakdown.
Week 8 (10/20-10/24)	FMEA analysis is completed and components are ranked by risk level.
Week 9 (10/27-10/31)	Continuing to work on IoT interface on AWS platform - Using IoTCore to send messages (input) - Receiving MQTT messages as client (output)
Week 10 (11/3-11/7)	Implementing GUI interface and try to get data from specific geographic regions Investigate use of ESP32 with sensors – meant to be transmission point between AWS and IoT
Week 11 (11/10-11/14)	The final modeling system is discussed and beginning to develop.
Week 12 (11/17-11/21)	Individual tasks are combined to develop the final modeling system.
Week 13 (11/24-11/28)	Work on Clinic Final Documents (Report and Presentation)
Week 14-15 (12/1-12/17)	Finalize Clinic Final Documents (Report and Presentation)

F. Engineering Standards

- IEEE 1451.0-2024: Standard for a Smart Transducer Interface for Sensors and Actuators [9]

The IEEE 1451.0-2024 standard defines a set of common interfaces and communication protocols for connecting sensors and actuators to measurement and control systems. It enables interoperability between sensors, embedded devices, and data acquisition platforms through standardized data formats and network interfaces. In the scope of this project, IEEE 1451 is relevant to the integration of vibration, temperature, and pressure sensors with embedded microcontrollers and IoT systems. Adhering to this standard supports reliable sensor communication, consistent data acquisition, and scalability of the predictive maintenance framework across multiple wind turbine assets.

- IEEE 2700-2017: Standard for Sensor Performance Parameter Definitions [10]

IEEE 2700-2017 establishes standardized definitions for sensor performance parameters such as accuracy, resolution, sensitivity, repeatability, and response time. These definitions are essential for evaluating sensor reliability and ensuring consistency in data interpretation. In this project, IEEE 2700 principles guide the selection and evaluation of condition-monitoring sensors used for predictive maintenance. Applying this standard helps ensure that collected sensor data accurately reflects turbine operating conditions and supports meaningful anomaly detection and prognostic analysis.

- ISO 13374-1:2003: Condition monitoring and diagnostics of machines – Data processing, communication and presentation [11]

ISO 13374-1:2003 provides a structured framework for condition monitoring systems, including data acquisition, signal processing, state detection, diagnostics, prognostics, and decision support. This standard is directly applicable to the predictive maintenance workflow implemented in this project, which follows a similar pipeline from raw sensor

and SCADA data collection to fault detection and maintenance recommendations. Compliance with ISO 13374 supports the systematic organization of monitoring data and ensures that predictive insights are actionable and aligned with industry best practices.

- ISO 55000:2024: Asset management – Vocabulary, overview and principles [12]

ISO 55000:2024 defines best practices for managing physical assets throughout their lifecycle to maximize value, reliability, and performance. Predictive maintenance is a core component of modern asset management strategies outlined in this standard. Within this project, ISO 55000 principles justify the transition from preventive to predictive maintenance by emphasizing risk-based decision-making, cost optimization, and lifecycle performance improvement for wind turbine assets.

V. Project Design

A. Problem Statement

Wind turbines experience continuous mechanical and environmental stresses that lead to gradual component degradation and unexpected failures. Current preventive maintenance approaches rely on fixed schedules and do not fully utilize available sensor and SCADA data, resulting in unnecessary maintenance and unplanned downtime. The design challenge of this project is to develop a predictive maintenance system that integrates real-time condition monitoring, data analytics, and risk prioritization to detect early anomalies and support proactive, data-driven maintenance decisions.

B. System Design Workflow

- Quantitative Data Analysis

The qualitative portion of this revolved around creating a relationship between components in a numerical manner. To do this, the SCATA data source was “cleaned,” meaning extreme outliers were removed so as not to skew the data in one direction. Then the team determined failure criteria as rotor speed at 0, and other indicators such as generator power production at 0 were identified. In order to better understand the components the failure indication that was determined to be most significant was the rotor rpm being 0. Then graphs were made comparing other components to RPM of the rotor. This gave us a y=mx+b relationship with an R^2 value. These relationships were then fed into the code to get an output of failure rate. The next steps in this process are to finish a MATLAB code that can pair the qualitative data and FEMA results with the existing relationships and output a failure rate that can then be fed into the code to create a predictive, accurate maintenance schedule.

- Qualitative Data Analysis

Qualitative data analysis (QDA) is the process of reviewing non-numerical information such as failure descriptions, maintenance logs, technician comments, OEM notes, and system behaviors to identify patterns, themes, and insights. In this case the QDA was used to understand why components fail, how they fail, and what these failures imply for overall system reliability. Instead of relying only on numeric metrics, the analysis focused on descriptive data, maintenance notes, observed failure modes, and component-level details found throughout the dataset.

1. *Systematic Coding and Categorization of failure modes*

All textual inputs were reviewed and manually coded according to the component category (e.g., rotor, pitch system, hub, gearbox, generator, nacelle systems).

For each component, failures entries were classified into distinct categories:

a. Failure mechanism (fatigue, erosion, corrosion, contamination, insulation breakdown, lubrication failure, etc.)
b. Immediate cause (e.g., pitting, delamination, bearing spalling, bolt preload loss)
c. Root cause (e.g., misalignment, moisture ingress, thermal cycling, inadequate lubrication, sensor drift)
d. Operational impact (efficiency loss, vibration increase, torque spikes, shutdown events)

This systematic categorization enabled direct compassion across components and highlighted cross-system reliability drivers.

2. *Thematic Pattern Recognition Across Subsystems*

Through iterative comparison of coded data, several high-level technical themes emerged:

a. Surface degradation processes: leading-edge erosion, gear tooth micropitting, brake disc glazing, and bearing race spalling.
b. contamination -driven failures: water-contaminated hydraulic oil, dust intrusion into slip rings, moisture ingress in bearings, and oil contamination in gearboxes.
c. Mechanical preload and alignment issues: bolt loosening in hub/pitch interfaces,

misalignment of main shaft and gearbox couplings, yaw bearing uneven loading.

d. Sensor and control-system degradation: encoder drift, LVDT signal loss, slip-ring communication noise, grounding faults.
e. Thermal and environmental stressors: overheating of motors and converters, cold start viscosity issues, salt-induced corrosion, lightning damage.

The recurring themes indicate that degradation mechanisms are not isolated events but systemic reliability risks tied to design, environmental exposure, and maintenance intervals.

3. Identification of highest-priority failure modes per component

a. Blades: Leading-edge erosion causing aerodynamic losses and reduced AEP.
b. Blade Root: Fatigue cracking at bolt interfaces and composite/adhesive joints.
c. Pitch System: Actuator malfunctions due to hydraulic contamination or electrical sensor drift.
d. Lightning Protection: Receptor and down-conductor degradation impacting strike dissipation.
e. Hub: Fatigue and bolt-preload-related failures at hub–blade flange interfaces.
f. Pitch Drives: Gear and bearing wear leading to backlash and pitch inaccuracy.
g. Main Shaft/Bearings: Pitting, brinelling, and lubrication breakdown increasing vibration and heat.
h. Gearbox: Gear micropitting and bearing failure—identified as primary drivers of major downtime.
i. Generator: Insulation degradation, fluting from shaft currents, slip-ring wear.
j. Yaw System: Motor overheating and bearing fretting from high-duty yaw cycles.
k. Brake System: Pad wear, pressure loss, and caliper sticking affecting overspeed protection.
l. Cooling System: Heat exchanger fouling and fan/pump failures causing thermal derating.

This step demonstrates how qualitative failure descriptions were converted into a technical hierarchy of risk.

4. Extraction of Dominant Maintenance Strategies

Using OEM maintenance instructions and recorded technician actions, the QDA identified the most critical maintenance techniques supporting reliability:

a. Lubrication health management: oil sampling, grease replenishment, flushing contaminated systems.
b. Structural Integrity Verification: bolt torque audits, flange inspections, NDT (UT, MPI, thermography)
c. Sensor calibration and electrical continuity: encoder recalibration, LVDT testing, slip ring cleaning, grounding checks.
d. Hydraulic system conditioning: filter replacement, accumulator pre-charge checks, leak detection.
e. Environmental and contamination control:dehumidification, seal inspections, desiccant breathers, SCADA-based temperature monitoring.

These maintenance activities were linked directly to the failure mechanisms identified, reinforcing the predictive maintenance value of the analysis.

5. *Technical interpretation of Reliability Implications*

a. Many critical failures originate from progressive degradation mechanisms not sudden faults.
b. Failures tend to propagate across subsystems (e.g., misalignment, bearing wear,gearbox damage, generator vibration). Environmental exposure (humidity, lightning, dust, turbulence) substantially accelerates failure modes.
c. Control-system reliability depends heavily on sensor integrity and electrical pathway continuity, making drift and contamination high-risk issues.
d. Predictive maintenance opportunities exist wherever failure modes produce early detectable signatures (vibration, temperature trends, oil contamination, torque fluctuations).

The QDA thus serves as the foundation for obstructing a data driven predictive maintenance strategy.

- FMEA Analysis

FMEA stands for failure mode and effects analysis. It aids in identifying and prioritizing which components of the wind turbine require immediate attention. A complete component breakdown was found in the previous section to determine how the components fail and their root cause failure. By utilizing the component breakdown found in the Qualitative Data Analysis, each component was ranked on a 10-point scale based on Severity, Occurrence, and Detection. Severity defines how important a failure is, occurrence defines how often a component will fail, and detection defines how difficult a failure is to detect. After each component is given their ranking in each category, the three numbers are then multiplied to determine the component's RPN (risk priority number). The RPN numbers are then ranked on a scale of low, medium, and high risk. This ranking system is used to group which components in their respective systems will fail most often and will have the most overall impact on the failure of the wind turbine itself. Below is the link to the FMEA analysis excel sheet with all of the informational charts given.

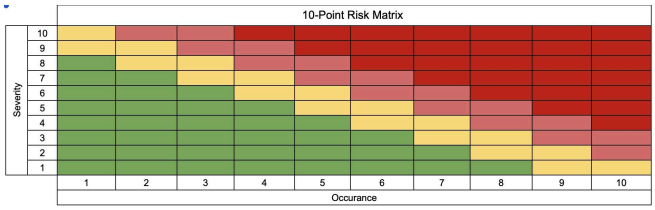


Figure 2: 10-Point Risk Matrix

The figure above is the color coded 10-point risk matrix that we designed to rank and organize the components.

Severity	Occurance	Detection
Minimal (1)	Very Remote (1)	Certain Detection (1)
Very Low (2)	Remote (2)	Very High (2)
Low (3)	Very Unlikely (3)	High (3)
Moderate (4)	Unlikely (4)	Moderately High (4)
Medium (5)	Possible (5)	Moderate (5)
Elevated (6)	Moderately Likely (6)	Moderately Low (6)
High (7)	Likely (7)	Low (7)
Very High (8)	Very Likely (8)	Very Low (8)
Severe (9)	Almost Certain (9)	Extemely Unlikely (9)
Critical (10)	Inevitable (10)	Impossible (10)

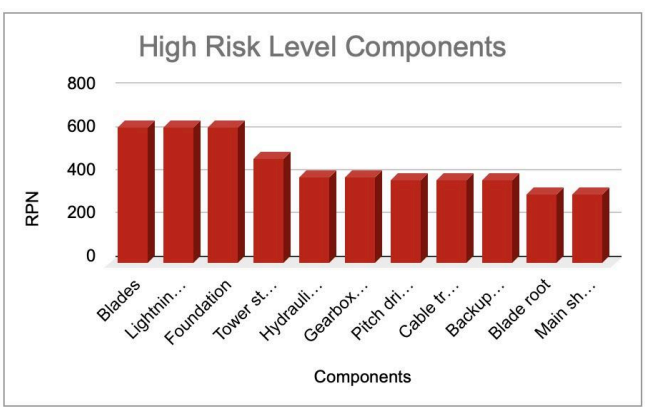
Figure 3: Determining Factors on 10-Point Scale

	Acceptable
	Consideration
	Need Corrective Action
	Needs Immediate Action

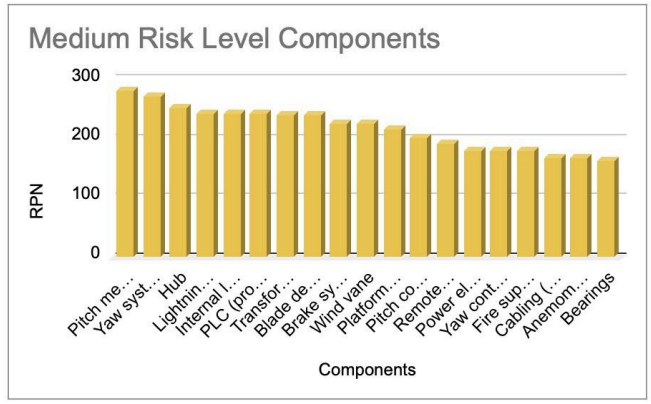
Figure 4: Determining Factors on 10-Point Scale

The two figures above are how the determining factors are broken down on a 10 point scale and what each color in the matrix is identifying.

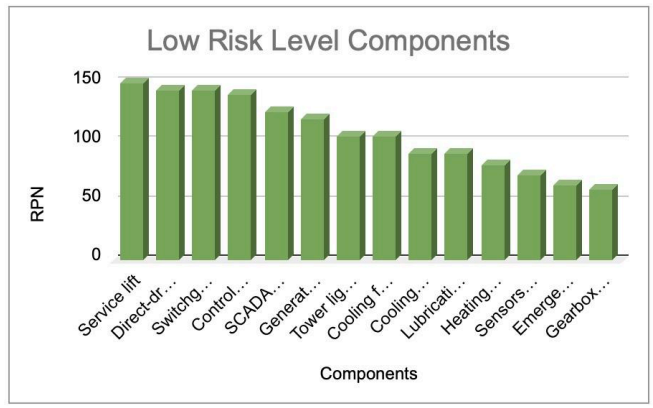
Graph A.



Graph B.

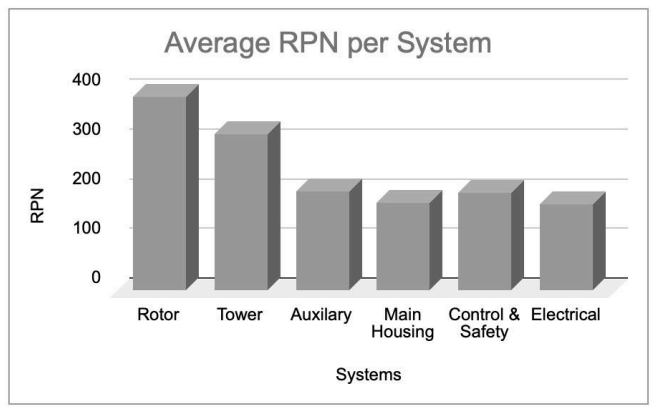


Graph C.



Graphs A, B, and C, are visual representations of each component's risk level. Since the risk levels are based off of RPN numbers, they are easily categorized in high, medium, and low risk.

Graph D.



Graph D is a visual representation of each system in the wind turbine's average RPN number. As shown, the rotor system has the highest average RPN and the electrical system has the lowest average RPN. This shows that a maintenance crew can focus on the higher ranking systems

first before diving deeper into the components of each system.

System	Components	Failure	Severity	Failure Cause	Root Cause	Occurance	Detection	RPN	Risk Level	Maintenance Procedures	Cost for Maintenance	Cost for replacement
Rotor System	Blades	Leading edge erosion	10	rain, hail, sand, and dust erode the blade's outer coating	High-speed rain/particle impact during normal operation, especially at blade tip speeds of 65–80 m/s	7	9	630	High	LEP renewal, composite repair, cleaning, erosion coating	$500–$40,000	Blade Replacement: $100,000–$250,000 per blade
	Blade root	Fatigue cracking in root bolts or laminates	10	Repeated cyclic loading from wind gusts causes micro-cracks in composite laminates or root bolts/flanges	High-cycle fatigue from turbulence, frequent start/stop cycles, and yaw misalignment	4	8	320	High	Bolt torque check, NDT, bolt replacement, laminate repair	$1,000–$40,000	Bolt Set Replacement: $8,000–$20,000
	Hub	Structural cracking of hub casting/welds	9	High cyclic bending loads from gusts or rotor imbalance produce micro-cracks in the cast or welded hub body	Casting porosity, residual welding stresses, or operating beyond design wind regime	4	7	252	Medium	NDT, weld repair, reinforcement, stress relief	$3,000–$60,000	Full Hub Replacement: $100,000–$250,000 + crane costs
	Pitch drives	Gear wear or pitting	8	Continuous oscillatory motion of drive gearbox during frequent pitch corrections	Inadequate or degraded grease/oil, misalignment of motor and ring gear, overload from abrupt braking or high turbulence	6	8	384	High	Oil change, gear inspection, gearbox replacement	$1,000–$60,000	Pitch motor replacement: $8,000–$20,000
Nacelle (Main Housing)	Gearbox (planetary, helical stages)	Gear tooth pitting / micropitting	10	High contact stress from continuous torque transmission	Inadequate lubrication film, misalignment of planetary or helical gears, contaminated oil increasing wear particles	4	5	400	High	Oil analysis, oil change, sensor installation, gear repair	$300–$40,000	Full Gearbox Replacement: $150,000–$350,000 + crane costs
	Gearbox lubrication system	Oil pump failure	4	Loss of oil flow or pressure	Pump motor burnout, bearing wear, cavitation from clogged suction strainer, low oil level, or air ingress	5	3	60	Low	Pump replacement, flush, filter change	$3,000–$8,000	Complete Replacement: $24,000

Figure 4: Correlation between SCADA data and FMEA Analysis

The figure above shows a direct correlation between the previously stated SCADA data and its respective component FMEA analysis. Since the data focused on the relationship between the rotor system and gearbox, those components were separated into a separate breakdown. As well as these component's failures, root causes of failures, RPN numbers, and risk levels, we have also added their respective maintenance procedures, cost of maintenance, and cost of replacements.

Overall, a FMEA analysis is critical in determining when a component is most likely to fail and how destructive that failure is to the overall machine. In our focus on wind turbines, we have identified several high risk components that will aid in a predictive maintenance plan for engineers. Also, with the SCADA data given and analyzed previously, we have focused on those components to develop a maintenance plan that will reduce the cost of general maintenance on the turbines.

- Sensor Condition Monitoring

Sensor condition monitoring forms the primary data acquisition layer of the predictive maintenance system by continuously capturing physical indicators of wind turbine health. In this project, vibration and temperature sensors are deployed to monitor critical subsystems such as the gearbox, bearings, and drivetrain. A vibration sensor (SW420) is used to detect abnormalities associated with bearing wear, imbalance, or shaft misalignment, with adjustable sensitivity and digital threshold-based output suitable for early fault detection. Temperature sensors are selected with an operating range sufficient for turbine environments and a resolution capable of detecting gradual overheating trends in gearboxes and rotating components. Sensor outputs are interfaced with an ESP32 microcontroller, which provides analog-to-digital conversion, GPIO connectivity, and wireless communication. Data is sampled at a rate appropriate for capturing dynamic mechanical behavior and transmitted via Wi-Fi to the cloud-based IoT platform. By establishing baseline operating thresholds and monitoring deviations over time, this sensor configuration enables reliable anomaly detection and supplies high-quality input data for downstream analytics, prognostic modeling, and maintenance decision support.

- IoT Interface

The AWS IoT interface functions as the secure communication and data management layer of the predictive maintenance system, enabling real-time transmission of sensor telemetry from the wind turbine to cloud-based analytics services. In this project, the ESP32 microcontroller connects to AWS IoT Core using the MQTT protocol over Wi-Fi, publishing JSON-formatted sensor data to designated topics at configurable sampling intervals. Device authentication is implemented using encrypted certificates to ensure secure, authenticated communication. AWS IoT Rules are configured to route incoming data to backend services such as DynamoDB for structured storage and AWS Lambda for event-driven processing. Lambda functions perform data validation and formatting and can export aggregated sensor data into CSV files for offline analysis, reporting, and model training. This design supports low-latency data delivery, scalable historical data storage, and seamless integration with data analytics tools, enabling continuous monitoring, anomaly detection, and long-term trend evaluation within the predictive maintenance framework.

- Data Processing

Data processing is a key part of our design process for this project. Once quantitative and qualitative data analysis are completed and combined with other project design aspects, we created a data processing model. This model was completed as a python code, and functions as a way to read and analyze large sets of scalar data. For our project scope, we used a large set of scalar data from sensors on a wind turbine that was input into an excel file. The code reads the data and uses the IQR, (Inter-Quartile-Range), method to identify outliers in the sensor data. Using these outlying data points the code then links the sensor names to recommended maintenance action items for possible affected components of the wind turbine. This is an extremely important part for predictive maintenance as it can recommend these action items based on data points before failures actually occur.

```
# 3. DETECT OUTLIERS USING IQR METHOD
# ============================================
def detect_outliers_iqr(data, column, multiplier=1.5):
    """
    Detect outliers using Interquartile Range (IQR) method
    multiplier: typically 1.5 for outliers, 3.0 for extreme
outliers
    """
    Q1 = data[column].quantile(0.25)
    Q3 = data[column].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - multiplier * IQR
    upper_bound = Q3 + multiplier * IQR

    outliers = (data[column] < lower_bound) | (data[column] >
upper_bound)

    return outliers, lower_bound, upper_bound

# Create a dataframe to store outlier flags
outlier_flags = pd.DataFrame(index=df.index)

# Dictionary to store outlier statistics
outlier_stats = {}
```

```python
print("="*60)
print("OUTLIER DETECTION RESULTS")
print("="*60)

for col in sensor_columns:
    # Skip if column has too many missing values
    if df[col].isna().sum() / len(df) > 0.5:
        print(f"\n⚠️  {col}: Skipped (>50% missing data)")
        continue

    # Detect outliers
    outliers, lower, upper = detect_outliers_iqr(df, col)
    outlier_flags[f'{col}_outlier'] = outliers

    # Count outliers
    num_outliers = outliers.sum()
    pct_outliers = (num_outliers / len(df)) * 100

    outlier_stats[col] = {
        'count': num_outliers,
        'percentage': pct_outliers,
        'lower_bound': lower,
        'upper_bound': upper
    }

    print(f"\n📊 {col}:")
    print(f"   Outliers: {num_outliers} ({pct_outliers:.2f}%)")
    print(f"   Valid range: [{lower:.2f}, {upper:.2f}]")

    # Display warning if outliers detected
    if num_outliers > 0:
        if pct_outliers > 5:
            print(f"   ⚠️  WARNING: HIGH outlier rate detected!")
        else:
            print(f"   ⚠️  WARNING: Outliers detected!")
```

Above is the most important part of the code design, which is the section which deals with the IQR analysis. This is the backbone in the code which identifies outliers and stores them to be categorized by severity. Although other data analysis methods could have been used, for the project scope IQR was the definitive choice.

As well as the IQR based code we also made a second code based upon equations created using relationships between sensors found through quantitative analysis. There were six equations utilized to each find specific failure types for the wind turbine. The code was able to find failure through these equations by using them to find an expected value for the sensors and then compare them to the actual values and then place the turbine as either healthy, pre-failure, or failure based on the deviation from the expected. The figure shown below helps to visualize how the code tracks turbines deviation from expected at each timestamp.

```python
y_pred = slope * x + intercept
r = (y - y_pred).abs()

mad = np.median(r)
if mad == 0 or np.isnan(mad):
    std = r.std()
    if std == 0 or np.isnan(std):
        # no variation at all -> no events, zero score
        idx = df.index
        return (
            pd.Series(False, index=idx),
            pd.Series(False, index=idx),
            pd.Series(0.0, index=idx),
        )
    mad = std

roll_dev = r.rolling(ROLL_WINDOW, min_periods=1).mean()
score = roll_dev / (mad + 1e-9)  # normalized deviation
```

The code in the first line is using the equations to find an expected value then under it finding a residual which is the difference between the expected and actual. Next the Median Absolute Deviation (MAD) is calculated and later used to score deviation. Then rolling deviation is found which is utilized in finding trends in the data and not just single spikes and finally then a deviation score is calculated using both the rolling deviation and MAD. The score is then later used to classify a timestamp as healthy, pre-failure, or failure. The final output of this code is the amount of timestamps that fall under each period as well as the sub-system each equation falls under and the specific failure that it finds in those systems.

The end goal with these two codes is to create a single code utilizing the advantages of both to create the best possible output to train an ai model off. The IQR based method is effective in seeing unusual sensor behavior in the short term. The equation based model in contrast uses real physical relationships between sensors which produces meaningful health and pre-failure labels. In a single code utilizing both the IQR outlier metrics are used as statistical anomaly features while the equation based code deviation score provides physics informed context. This allows the ai to learn both abnormal behavior and true mechanical degradation for more accurate predictive maintenance.

VI. Results and Observations

A. Experimental/Simulation Results

1) Data Processing

In Figure 2, the output of our python code created in our coding environment, as well as explanations for each output and its significance, can be seen.

```
Scalar Data Failure Detection Code
File  Edit  View  Insert  Runtime  Tools  Help
Commands  + Code  + Text  Run all

============================================================
SENSORS RANKED BY OUTLIER COUNT
============================================================
1. generator speed (rpm).2            -  3046 outliers (17.76%)
2. pitch angle for blade 1 (deg)      -  2747 outliers (16.02%)
3. generator speed (rpm)              -  2215 outliers (12.91%)
4. rotor speed (rpm)                  -  2213 outliers (12.90%)
5. power factor                       -  2193 outliers (12.79%)
6. generator speed (rpm).1            -  1429 outliers ( 8.33%)
7. rotor speed (rs)                   -  1420 outliers ( 8.28%)
8. total operating hours of yaw       -   641 outliers ( 3.74%)
9. gearbox oil temperature (c)        -   465 outliers ( 2.71%)
10. operating frequency (hz)          -   297 outliers ( 1.73%)
```

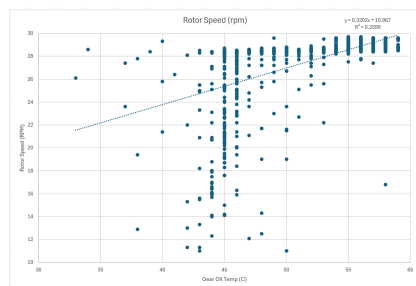
Above is the initial ranking produced by the IQR analysis, it detects the outliers by column heading in the file it is reading and categorizes them by the most affected at the top down to the least affected.

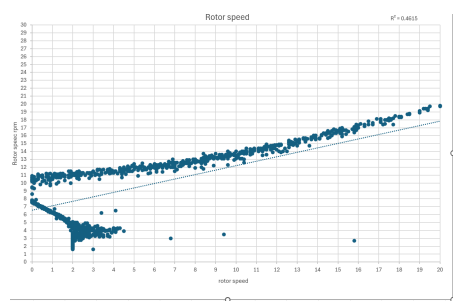
```
Scalar Data Failure Detection Code
File  Edit  View  Insert  Runtime  Tools  Help
Commands  + Code  + Text  Run all

============================================================
RECOMMENDED ACTION ITEMS
============================================================

Found 19 sensors with outliers. Recommended actions:


============================================================
🔧 GENERATOR SPEED (RPM).2
   Outliers: 3046 (17.76%)

   Action Items:
   1. 🚨 URGENT: Schedule immediate inspection
   2. Inspect mechanical connections and couplings
   3. Check for vibration or imbalance issues
   4. Verify sensor calibration and mounting

============================================================
🔧 PITCH ANGLE FOR BLADE 1 (DEG)
   Outliers: 2747 (16.02%)

   Action Items:
   1. 🚨 URGENT: Schedule immediate inspection
   2. Inspect pitch control system and actuators
   3. Check hydraulic pressure and fluid levels
   4. Verify pitch angle sensor calibration
   5. Examine blade bearings and pitch motors

============================================================
```

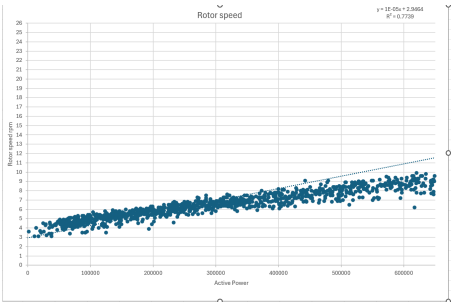
This section of the code is very important, shown is the top two affected data sets. In this case, it was generator speed and the pitch angle of the turbine blade. This part of the code is key for our predictive maintenance analysis. The code uses key words in the column headings to determine what physical components may be affected. Based on the identification of the physical components, the code then suggests maintenance action items. This can further reduce downtime of components and lessen cost for the maintenance as well as reduced power production due to this downtime. This was a large objective that we wanted to overcome, actually having recommended action items based on the affected sensors.

```
Equation: Gear oil temp vs rotor speed (gear_oil_vs_rotor)
  Primary subsystem : Gearbox
  What it detects   : Overheating, lubrication failure
  Failures flagged  : 2406 rows (14.03% of dataset)
  Prefailures flags : 3196 rows (18.63% of dataset)

Equation: Power vs stator temperature (stator_temp_vs_power)
  Primary subsystem : Generator
  What it detects   : Cooling failure, winding heating
  Failures flagged  : 297 rows (1.73% of dataset)
  Prefailures flags : 1168 rows (6.81% of dataset)

Equation: Hydraulic temp vs rotor speed (hydraulic_temp_vs_rotor)
  Primary subsystem : Pitch Actuator
  What it detects   : Pitch system inefficiency, overheating
  Failures flagged  : 1435 rows (8.37% of dataset)
  Prefailures flags : 1522 rows (8.87% of dataset)
```

Above is the output for the equation-based code. The figure shows the output for three of the six equations. As can be seen for each equation the code outputs the subsystem the equations relate to the failure types it can indicate as well as the amount of failures and pre-failures detected in the scada data. This output is very important for helping to develop an ai for predictive maintenance in its ability to be accurate and informative. With addition of the subsystem and failure type detection it can provide necessary insight for what needs service when ai flags pre-failure in the turbine. As well as this pre-failure flagging can help the ai to learn trends leading to failure allowing earlier and more accurate detection.







Above are the main graphs used as input for the code. Using excel we are able to derive and equation that tells us the slope and y intercept of the relationship that can then be used to find failure occurrences and premature failures. The regfinement of this process in order to input more digestable information to the code is ongoing. With that being said, the code is able to untepret the equations in a meaningful way and output the aforementioned metrics.

VII. Impact Statement

A critical analysis of the results are reflected by the below impact statements. Each statement encompasses design decisions made when configuring this predictive maintenance system, as well as overall impacts that are encompassed in general by implementing this type system on wind turbine applications.

A. Contribution to Knowledge

The Asset Predictive Maintenance clinic project contributes to the growing body of knowledge in predictive maintenance and renewable energy systems by demonstrating an integrated, data-driven system for wind turbine health monitoring. This project combines sensor-based condition monitoring, SCADA data analysis, machine learning–based anomaly detection, and Failure Mode and Effects Analysis (FMEA) into a unified predictive maintenance approach. By linking quantitative analytics with qualitative risk prioritization, the system illustrates how raw operational data can be transformed into actionable maintenance insights rather than isolated diagnostic outputs.

Additionally, this work highlights the practical application of IoT infrastructure and cloud-based analytics for scalable asset monitoring in renewable energy environments. The project reinforces the value of combining anomaly detection and prognostic modeling to improve early fault identification and maintenance planning. Overall, the framework developed in this project serves as a reference model for implementing predictive maintenance strategies in wind turbines and similar industrial assets, contributing to both academic understanding and practical engineering implementation.

B. Public health, Safety and Welfare factors

For the Asset Predictive Maintenance clinic project, an IoT-based predictive maintenance system was designed to improve the technical safety and reliability for wind turbines. In order to reduce the likelihood of mechanical failures, it was crucial to enable early detection of the overheating of components, increased vibration, and structural abnormalities. By developing a system that is able to monitor the health of a wind turbine in real time and provide alerts, it can be recognized how this approach directly enhances public safety by preventing hazardous operating conditions, reducing the chance of unexpected turbine breakdowns, and allowing for proper maintenance through predicting potential component failure before unsafe conditions escalate.

C. Global, Social, Cultural factors

The Asset Predictive Maintenance clinic project highlighted how wind turbines are fundamental to energy generation from a global standpoint. It was necessary for this predictive maintenance solution to be adaptable to diverse geographical, social, and cultural environments that these wind turbines would operate in. Designing a system that is cost-effective and can be easily implemented, this project was able to support the adoption of renewable methods of energy generation and ensure that those of varying geographic regions are accessible to reliable operation of wind turbines regardless of external factors like infrastructure.

D. Environmental factors

This clinic project supported environmental sustainability as the creation of a system that predicts and mitigates the chances of component failure will minimize the amount of waste produced from premature maintenance and improve the operational lifespan of wind turbines. Monitoring indicators, like temperature and vibration, assists with the detection of degrading components of a wind turbine and lowers the larger-scale damage that could require increased repair or complete replacement. Taking environmental factors such as humidity and corrosion into consideration is crucial as they directly affect the health of the wind turbine. These factors must be incorporated to reflect true operating conditions to reduce the need for overconsumption of materials and repairs.

E. Economic factors

By aiming to reduce the operational costs that come with component reparations and the downtime of wind turbines, the Asset Predictive Maintenance clinic project has taken economic feasibility into account. Identifying abnormalities early on in the maintenance process will allow for the avoidance of larger failures that can ultimately lead to more expensive repairs and degradation to the overall lifespan of the turbine. Using cost-effective technology, like sensors and cloud services, ensures financial feasibility for all users.

This clinic project further supported economic factors by ensuring that maintenance is performed when data indicates increased chance of a high-cost emergency.

VIII. Ethical Considerations

In regard to global, economic, environmental, and societal factors, there were several ethical considerations that had to be made for this clinic project. When collecting data from sensors and sending it to the cloud (AWS), there was a concern about data accuracy as misleading information could lead to improper maintenance. It was important to ensure that abnormalities were being identified to minimize potential discrepancies that could lead to unreliable analysis. An additional ethical consideration that can be made is the economic accessibility of this maintenance system where a solution that is high functioning could be too expensive to be implemented in regions that are unable to afford it and require it the most for grid stability. By addressing these potential issues, this project was able to ensure safety, equity, and more environmentally-aware energy solutions.

IX. Discussion and Conclusion

The results of this project demonstrate the effectiveness of integrating sensor-based condition monitoring, IoT infrastructure, and data-driven analytics for predictive maintenance of wind turbines. The system successfully identified early warning indicators and high-risk components, particularly within the rotor and gearbox subsystems. Compared to traditional preventive maintenance approaches, the predictive framework provides earlier fault detection and more targeted maintenance recommendations.

While the project was limited by the availability of turbine-specific real-time data, the developed framework validates the feasibility of applying predictive maintenance strategies in wind energy systems. Overall, the project achieved its primary objectives by delivering a scalable, data-driven maintenance model that improves reliability, reduces operational costs, and supports proactive asset management in renewable energy applications.

X. Future Work

While the objective of developing a predictive maintenance framework for wind turbines was achieved, several opportunities exist to further enhance the system with additional time and resources. One primary area of future work involves strengthening the artificial intelligence model by expanding the size and diversity of training datasets and incorporating long-term historical SCADA and sensor data. This would improve model generalization, reduce false positives, and allow for more accurate prediction of remaining useful life across varying operating conditions.

Another important area for future development is adapting the predictive maintenance framework to specific wind turbine models and configurations. Different turbine manufacturers, rated power classes, and environmental operating conditions exhibit unique failure behaviors and degradation patterns. By calibrating the AI model to turbine-specific parameters, such as drivetrain design, control strategies, and site location, the system can deliver more precise maintenance insights. Establishing turbine-specific baselines and risk thresholds would further improve prediction accuracy and scalability, enabling the framework to be deployed effectively across diverse wind farm assets.

XI. Acknowledgment

We would like to give a special thanks to Dr. Yashwant Sinha for lending their knowledge and expertise relating to this project and guidance in data collection.

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